Hossein Fathollahian

Data visualization/Scientist/AI Engineer

Computer Science PhD Student
Electronic Visualization Laboratory
University of Illinois Chicago, IL
hfatho2@uic.edu

github.com/Hosseinfatho
linkedin.com/in/hossein-fathollahian
Portfolio: hosseinfatho.github.io
+1 (312) 498-3674

My research focuses on visual analytics for high-dimensional biological and biomedical imaging data, leveraging AI, computer vision, machine learning, and human-centered design to transform complex data into actionable insights. I seek to bridge AI and biomedical discovery through intelligent visualization platforms that help biologists and clinicians explore disease processes, support diagnosis and treatment decisions, and advance drug discovery and delivery.

Skills

Programming Languages	Python, JavaScript, MATLAB, C++
Frameworks/Libraries	PyTorch, TensorFlow, Keras, OpenCV, Pandas, NumPy, SciPy, scikit-learn, Dask, Matplotlib, Plotly
Developer Tools	PyTorch, TensorFlow, Keras, OpenCV, Pandas, NumPy, SciPy, scikit-learn, Dask, Matplotlib, Plotly
Communication	English, Farsi (fluent speaker)

Technical Experience

Researcher / ConGAT (3D Region of Interest Discovery in Multiplexed Microscopy Images) **Nov, 2025 — Present**
UIC(EVL) (Prof. Saeed Boorboor), (Prof. G. Elisabeta Marai) *Chicago, IL*

- Self-supervised graph attention for spatial protein interaction analysis.
- Visual computing techniques for early-stage and low-intensity protein interaction detection.
- Interactive 3D visual analytics for biomarker exploration and hypothesis generation.

Research Collaboration, Visual analytics / BioSet (Interactive Visual Analysis of Biomarker Co-localization) **Dec, 2025 — Present**
UIC(EVL)-NYU (Prof. G. Elisabeta Marai),(Prof. Saeed Boorboor),(Prof.Robert Krueger) *Chicago, IL*

- Automated entropy-driven camera positioning for optimal visual exploration.
- Provenance-aware bookmark lineage and exploration history management.
- Seamless high-resolution image navigation using computer vision techniques.
- Agentic AI-powered report generation and visual storytelling from annotations.

Researcher, Visualization / Loom (Multi-Region Analysis of Spatial Transcriptomics) **May, 2025 — Present**
UIC(EVL) (Prof. G. Elisabeta Marai) *Chicago, IL*

- Developed trajectory analysis and visualization methods for spatial transcriptomics exploration.
- Designed and refined glyph-based visual encodings for cellular and spatial pattern analysis.
- Implemented multi-resolution and clustering visualizations for scalable exploration of large datasets.

Researcher, Computational Social Science / AI and Visual Analytics for Digital Communication **Aug, 2025 — May, 2026**
UIC(Communication department)(Prof. Zizi Papacharissi) *Chicago, IL*

- Built data collection and preprocessing pipelines for large-scale social media datasets.
- Developed visual analytics tools to explore communication patterns, engagement, and online discourse.
- Applied AI and machine learning methods to analyze human communication behaviors across digital platforms.

Data Scientist and Visualization Engineer / CAIDF (Creating AI-Enabled All-Health Team Data Fabric) **Feb, 2025 — Sep, 2026**
UIC(Associate Vice Chancellor for Research and Chief Research Information Officer)(Prof. Andrew D. Boyd,) *Chicago, IL*

- Developed a visual analytics dashboard integrating clinical histories, NLP results, and cohort patterns.
- Designed interactive visualizations for longitudinal analysis and patient–cohort comparisons.
- Collaborated with clinicians and NLP researchers to visualize and interpret complex healthcare data.

Visualization-AI Engineer / SAGE (The Edge Computing for Wildfire Detection/ Healthcare/ Discovery) **May, 2025 — Aug, 2025**
Argonne National Laboratory-(EVL)(Prof. Michael E. Papka) *Chicago, IL*

- Edge AI pipelines for real-time sensor data analytics.
- Sensor integration and machine learning workflow development.
- Real-time ML model deployment on NVIDIA Jetson devices.

Software developer / BI-LAVA (Biocuration with Hierarchical Image Labeling) **Jan, 2024 — Aug, 2024**
UIC(EVL) (Prof. G. Elisabeta Marai) *Chicago, IL*

- Implemented front-end features for taxonomy exploration, image visualization, and active learning workflows.
- Enhanced backend architecture and Docker-based DevOps infrastructure for reliable deployment and development.
- Accelerated machine learning search and data processing pipelines for large biomedical image repositories.

Teaching Assistant / C++ (CS211-CS141: Computer Programming practicum) **Jan, 2024 — May, 2025**
UIC (Prof. Scott Reckinger), (Prof. Mark Hodges) *Chicago, IL*

- Conducted lab sessions and facilitated active learning in introductory computer science courses.
- Provided programming support and mentorship in C++, debugging, and software development fundamentals.
- Assessed assignments and examinations while promoting student engagement and learning outcomes.

Education

PhD Student in Computer Science, *University of Illinois Chicago (UIC)* 2024-2028
Master of Telecommunication Engineering, *Shahid Beheshti University* 2010-2012
Bachelor of Electronics Engineering, *Semnan University* 2006-2010

Activities

Awards - Honors

*** Patricia Flatley Brennan Award for Informatics Team Science, American Medical Informatics Association (AMIA)**:
"Recipient as a member of the CAIDF research team (Creating AI-Enabled All-Health Team Data Fabric)". Nov, 2026
*** Best Poster Award – ESSB (European Society for Spatial Biology Conference) Sitges, Spain**:
"Scalable 3D Visualization and Spatial Exploration of Multiplexed Cancer Tissue Images for the Web". Oct, 2026
*** Award of Merit – IEEE VIS Bio+MedVis Challenge:**
"Attention-Based ROI Discovery in 3D Tissue Images" Aug, 2025
*** Award of Merit – IEEE VIS Bio+MedVis Challenge:**
"From Peaks to Patterns" Aug, 2025

Professional Service

*** Peer Review Service:**
IEEE VIS, CHI, DIS, IDC, IMX, CogSci, AMCIS 2024–2026
*** IEEE VIS Student Volunteer:** *IEEE Visualization & Visual Analytics Conference in Boston, USA* Nov, 2026

Publications

1. **Fathollahian, H.**, Zhao, S., Nipu, N., Salahshour, M., Duan, L., Tang, W., BoorBoor, S., and Marai, G. E. “ConGAT: A Context-Aware Graph Attention Network for Region-of-Interest Discovery in Multichannel Microscopy Environments.” *Computers & Graphics* (2026). https://doi.org/10.1016/j.cag.2026.104728.

2. Zhao, S., Nipu, N., **Fathollahian, H.**, Salahudeen, A., Chen, H., Karginova, O., and Marai, G. E. “Loom: Multi-Region Analysis of Spatial Transcriptomics with Local Neighborhoods and Global Trajectories.” *IEEE Transactions on Visualization and Computer Graphics* (2026).

3. **Fathollahian, H.**, Zhao, S., Salahshour, M., BoorBoor, S., and Marai, G. E. “ConformaLens: Conformational States in Adaptive Molecular Dynamics.” *IEEE Transactions on Visualization and Computer Graphics* (Bio+MedVis Challenge 2026) (2026): 12–14.

4. Tayebi, S., Oti, A. S., **Fathollahian, H.**, and Haque, U. “Mapping War Trauma: A Machine Learning Approach to Predict Mental Health Impacts in Ukraine.” *SSM–Population Health* (2025): 101879. https://doi.org/10.1016/j.ssmph.2025.101879.

5. **Fathollahian, H.**, Zhao, S., Nipu, N., and Marai, G. E. “Attention-Based ROI Discovery in 3D Tissue Images.” *IEEE Transactions on Visualization and Computer Graphics* (Bio+MedVis Challenge 2025) (2025): 2–4.

6. **Fathollahian, H.**, and Salahshour, M. “From Peaks to Patterns.” *IEEE Transactions on Visualization and Computer Graphics* (Bio+MedVis Challenge 2025) (2025): 22–24.

7. Torkamani-Azar, F., Imani, H., and **Fathollahian, H.** “Video Quality Measurement Based on 3-D Singular Value Decomposition.” *Journal of Visual Communication and Image Representation* 27 (2015): 1–6. https://doi.org/10.1016/j.jvcir.2014.12.004.